

Focused literature review for inverse Markov-chain route-choice inference

My short, opinionated read is that the strongest adjacent literatures for your pipeline are not generic Markov estimation papers, but the information-geometry literature on Markov kernels, the recursive-logit route-choice literature, and Bayesian smoothing work on sparse transition matrices. The first speaks directly to the optimisation pathology you are seeing with L-BFGS-B; the second is the cleanest one-step-Markov way I found to close the misspecification gap you report against the empirical one-step ceiling; and the third gives a principled replacement for the hard floor in your sparse g -table. On the detection side, the most useful result is that mixture/segmentation tests for Markov models have non-regular asymptotics, so any “extra component” or “switching behaviour” score should be calibrated by bootstrap or posterior predictive methods rather than naïve χ^2 arguments. On the transport side, the literature is very clear that “all app users minimise congestion” is too strong; partial adoption, imperfect information, bounded rationality, and mixed equilibria are the standard formulations. ¹

Optimisation

Papers I would keep closest to hand. Masahito Hayashi and Shun Watanabe, “Information geometry approach to parameter estimation in Markov chains,” *Annals of Statistics* 44(4), 2016, DOI 10.1214/15-AOS1420. This is the cleanest theoretical justification I found for Fisher-geometry-aware fitting on exponential families of transition matrices. ²

Masahito Hayashi, “Information Geometry Approach to Parameter Estimation in Hidden Markov Models,” *Bernoulli* 28(1), 2022; *arXiv:1705.06040*. This is especially relevant because it treats partial observation and shows how estimation error is governed by a projective Fisher information, not the naïve Euclidean geometry of the parameter vector. ³

Mark K. Transtrum and James P. Sethna, “Improvements to the Levenberg–Marquardt algorithm for nonlinear least-squares minimization,” *arXiv:1201.5885*, 2012. This is not Markov-specific, but your objective is exactly a weighted nonlinear least-squares problem in log-residuals, and this paper is unusually concrete about robustness to narrow canyons, poor conditioning, and sensitivity to starting points. ⁴

The concrete mapping is straightforward. Hayashi–Watanabe says your current substitution of L-BFGS-B is not merely a taste issue; it changes the geometry of the problem from the Fisher metric on transition probabilities to an arbitrary Euclidean geometry on softmax parameters. In your setup, that makes the paper’s original warning about bad basins unsurprising rather than incidental. The first thing I would implement is a blockwise natural-gradient or Fisher-scoring step for each state’s outgoing softmax, with either exact rowwise Fisher blocks or a cheap diagonal-plus-low-rank approximation. The second thing I would test is a damped Gauss–Newton / Levenberg–Marquardt solver on the residual vector $(\log \hat{f} - \log f_\theta, \log \hat{g} - \log g_\theta)$, because your loss already has the right structure for it. A hybrid would be strongest: Gauss–Newton in residual space, preconditioned by the Fisher blocks of the softmax parameterisation. ⁵

My implementation judgement is: full natural gradient or Fisher scoring is **medium** difficulty and **high** expected payoff; geodesic LM is **medium** difficulty and **medium-to-high** payoff; generic L-BFGS-B

variants are **low** difficulty and **low** expected payoff. I did not retrieve a published method more convincing for your exact objective than “restore the Markov/Fisher geometry” plus “use a least-squares-aware second-order method”; I would not spend a month benchmarking Adam, RMSProp, or generic quasi-Newton tweaks before doing those two. ⁵

Cross-cutting point: Hayashi’s partial-observation work is not only an optimisation pointer; it also gives the right language for uncertainty and identifiability in topic **G**, because the projective Fisher information tells you which directions are actually estimable from your observation operator. ⁶

Richer first-order model families

Papers I would keep closest to hand. *Mogens Fosgerau, Emma Frejinger, and Anders Karlström, “A link based network route choice model with unrestricted choice set,” Transportation Research Part B 56, 70–80, 2013, DOI 10.1016/j.trb.2013.07.012.* This is the single most relevant model-family paper I found for your misspecification problem: it keeps route choice one-step Markov at the link level, but adds a downstream value function so local choice probabilities depend on destination and network continuation value. ⁷

Tien Mai, Mogens Fosgerau, and Emma Frejinger, “A nested recursive logit model for route choice analysis,” Transportation Research Part B 75, 100–112, 2015, DOI 10.1016/j.trb.2015.03.015. This is the natural next step when simple recursive logit still underfits because overlapping routes and substitution patterns violate IIA. ⁸

Tim P. van Oijen, Winnie Daamen, and Serge P. Hoogendoorn, “Estimation of a recursive link-based logit model and link flows in a sensor equipped network,” Transportation Research Part B 140, 262–281, 2020, DOI 10.1016/j.trb.2020.08.003. This is the best direct bridge to your observation regime: it estimates a recursive link-based logit model from gappy, partial sensor observations rather than complete paths. ⁹

Rishi Gupta, Ravi Kumar, and Sergei Vassilvitskii, “On Mixtures of Markov Chains,” NeurIPS 29, 2016; and Chinmaya Kausik, Kevin Tan, and Ambuj Tewari, “Learning Mixtures of Markov Chains and MDPs,” ICML / PMLR 202:15970–16017, 2023. These two papers are the best latent-heterogeneity references I found for keeping the model first-order while allowing multiple behavioural regimes. ¹⁰

For your problem, recursive logit is the standout. Your current local softmax can only express “which outgoing edge looks good right now”. Recursive logit expresses “which outgoing edge looks good once I account for the best expected utility to the destination”, which is exactly the kind of network look-ahead that can explain why your empirical one-step ceiling sits well above the fitted softmax family. If you know trip destinations in the simulator, I would make destination part of the conditioning context immediately. If you only know it late in the path, I would fit destination-conditioned chains on training data and score each test trajectory by integrating or maximising over candidate destinations. Then, if route overlap still matters, move to nested recursive logit; if driver heterogeneity matters, add a small latent mixture over the recursive-logit utilities rather than abandoning the first-order structure. ¹¹

My implementation judgement is: basic destination-conditioned recursive logit is **medium** difficulty and **very high** expected payoff; nested recursive logit is **high** difficulty and **medium-to-high** payoff; a small latent mixture of first-order chains is **medium-to-high** difficulty and **high** payoff if your “intrinsic versus sat-nav” classes are themselves heterogeneous. I would explicitly *deprioritise* GNNs here. I did not retrieve a graph-neural paper that matched your exact constraint set as cleanly as recursive-logit plus

mixture models, and the interpretability and partial-observation story are both materially better in the recursive-logit literature. ¹¹

Cross-cutting point: van Oijen et al. is also important for topic **F**, because it shows that route-choice estimation under partial sensing can be validated against network flows, not only against path likelihood. Gupta and Kausik are also central for topic **D**, because they naturally lead to trajectory-level classification rules rather than only point-estimate LLR tests. ¹²

Bayesian shrinkage for sparse hitting rates

Papers I would keep closest to hand. *Yousif Alyousifi, Kamarulzaman Ibrahim, Wei Kang, and Wan Zawiah Wan Zin, "Robust empirical Bayes approach for Markov chain modeling of air pollution index," Journal of Environmental Health Science and Engineering 19(1), 343–356, 2021, DOI 10.1007/s40201-020-00607-4.* This is the most directly useful "zero-frequency smoothing in sparse Markov count matrices" paper I found. ¹³

Yousif Alyousifi, Kamarulzaman Ibrahim, Wei Kang, and Wan Zawiah Wan Zin, "Markov chain modeling for air pollution index based on maximum a posteriori method," Air Quality, Atmosphere & Health 12, 1521–1531, 2019, DOI 10.1007/s11869-019-00764-y. This is practically useful because it compares Dirichlet, Jeffreys, and uniform priors for sparse transition estimation and reports precision gains over pure MLE in small samples. ¹⁴

Souad Assoudou and Belkheir Essebbbar, "A Bayesian Model for Markov Chains via Jeffrey's Prior," Communications in Statistics — Theory and Methods 32(11), 2163–2184, 2003, DOI 10.1081/STA-120024474. This is the most relevant older paper I found for noninformative Bayesian smoothing that allows correlation structure in transition probabilities rather than independent rowwise pseudocount thinking. ¹⁵

The practical lesson is simple: your 10^{-3} floor is acting as an implicit pseudo-likelihood that overwhelms the small amount of real information in g . The literature points to replacing that with shrinkage, not a floor. For your setup I would use a two-level procedure. First, define binary or count data for "from observed state i , did the trajectory later hit observed state j before termination / before leaving the considered segment?" and apply a Beta-Binomial or Dirichlet empirical-Bayes smoother, with hyperparameters learned across all (i, j) cells or across feature-sharing groups such as topological distance, district, or road class. Second, if you want full coherence with the fitted chain, place the prior on P , sample or approximate its posterior, solve the linear hitting-probability equations for each posterior draw, and plug the posterior mean or a posterior predictive score for g into the loss instead of a hard floor. The papers do not implement that exact last step, but they make it the natural Bayesian extension. ¹⁶

My implementation judgement is: empirical-Bayes smoothing on g -cells is **low** difficulty and **medium** payoff; posterior propagation from P to g is **medium** difficulty and **medium-to-high** payoff if you insist on keeping g in the objective. The critical point is that g should either be weakly regularised auxiliary evidence or posterior-integrated evidence; it should not be a matrix of mostly artificial floor values that contributes loss mass out of proportion to its information content. ¹⁶

Cross-cutting point: these same priors are also the cleanest route into topic **G**. Once you have a posterior on P , you can report uncertainties on downstream detection scores, not just on rows of the transition matrix. ¹⁷

Trajectory detection and scoring

Papers I would keep closest to hand. *Matthew Fitzpatrick and Michael I. Stewart, "Asymptotics for Markov chain mixture detection," Econometrics and Statistics 22, 56–66, 2022, DOI 10.1016/j.ecosta.2021.11.004.* This is the sharpest warning I found for your detection layer: likelihood-ratio tests involving Markov mixtures have non-standard asymptotics, so standard χ^2 calibration is wrong. ¹⁸

Petter Arnesen, Tracy Holsclaw, and Padhraic Smyth, "Bayesian Detection of Changepoints in Finite-State Markov Chains for Multiple Sequences," arXiv:1504.00860, 2015. This is the best match if rerouting is not a trip-level latent label but a within-trip regime switch. ¹⁹

Chinmaya Kausik, Kevin Tan, and Ambuj Tewari, "Learning Mixtures of Markov Chains and MDPs," ICML / PMLR 202:15970–16017, 2023. This gives a concrete trajectory-clustering and classification pipeline from short unlabeled sequences and is highly relevant if your "intrinsic" and "sat-nav" classes are not cleanly prelabelled at training time. ²⁰

Michael Görnitz, Nico S. Görnitz, Malte Kloft, and Ulf Brefeld, "Hidden Markov Anomaly Detection," PMLR 37, 2015. This is not route-choice-specific, but it is the best anomaly-scoring paper I retrieved that explicitly respects sequence structure rather than flattening each trip into i.i.d. features. ²¹

For your detector, the most useful split is this. If the behavioural label is whole-trip, use a model-averaged predictive density rather than a single-chain LLR: average the per-trajectory predictive probabilities over several optimiser basins or latent components, then score with the log posterior odds or log predictive density ratio. If rerouting may start mid-trip, your current score is misspecified and a changepoint Markov model is the right baseline comparator. If labelled anomalies are scarce or unreliable, use a one-class structured detector trained on "intrinsic" trajectories only, with the sat-nav trajectories treated as deviations from nominal behaviour rather than a fully specified second class. ²²

My implementation judgement is: model-averaged LLRs over several fitted basins are **low-to-medium** difficulty and **medium** payoff; changepoint scoring is **medium** difficulty and **high** payoff if rerouting is episodic within a journey; mixture-of-Markov-chain classification is **medium** difficulty and **medium-to-high** payoff when behavioural heterogeneity inside each class is large. The one thing I would not do is interpret any "one component versus two components" LRT in this space with regular asymptotics; Fitzpatrick–Stewart shows that this is exactly the regime where that shortcut fails. ²³

Cross-cutting point: Kausik and Gupta belong equally to topic **B** as a model-family extension, and Arnesen's posterior predictive machinery belongs equally to topic **G** as an uncertainty-quantification strategy for downstream scoring. ²⁴

Sat-nav realism and route-choice fixed points

Papers I would keep closest to hand. *J. G. Wardrop, "Some Theoretical Aspects of Road Traffic Research," Proceedings of the Institution of Civil Engineers 1(3), 325–362, 1952, DOI 10.1680/ipeds.1952.11259.* This is still the foundational equilibrium reference: selfish users equalise perceived costs, which is different from network-wide congestion minimisation. ²⁵

Hai Yang and Qiang Meng, "Modeling user adoption of advanced traveler information systems: dynamic evolution and stationary equilibrium," Transportation Research Part A 35(10), 895–912, 2001, DOI 10.1016/S0965-8564(00)00030-6. This is the most directly relevant paper I found on heterogeneous adoption of route-guidance information. ²⁶

David Watling and Martin L. Hazelton, "The Dynamics and Equilibria of Day-to-Day Assignment Models," *Networks and Spatial Economics* 3, 349–370, 2003, DOI 10.1023/A:1025398302560. This is the best general reference I found for why route choice under congestion should often be formulated as a day-to-day dynamical system rather than a single one-shot fixed point. ²⁷

Xiaomei Zhao, Chunhua Wan, and Jun Bi, "Day-to-Day Assignment Models and Traffic Dynamics Under Information Provision," *Networks and Spatial Economics* 19, 473–502, 2019, DOI 10.1007/s11067-018-9386-1; and Saeed Asadi Bagloee, Majid Sarvi, Michael Patriksson, and Abbas Rajabifard, "A Mixed User-Equilibrium and System-Optimal Traffic Flow for Connected Vehicles Stated as a Complementarity Problem," *Computer-Aided Civil and Infrastructure Engineering* 32(7), 562–580, 2017, DOI 10.1111/mice.12261. Together these are the most useful papers I found for modelling informed and uninformed users, and for representing "guided" users as a separate class rather than assuming everyone solves the same optimisation problem. ²⁸

The literature says your present sat-nav behavioural story is directionally plausible but too strong. Real route-guidance users are usually modelled as a partially adopting, heterogeneous subgroup responding to information that is imperfect, delayed, and filtered through bounded rationality. In other words, the realistic comparator to your intrinsic chain is not a "system-optimal chain" for all rerouting users, but a *mixed* system in which some users follow selfish user-equilibrium or stochastic user-equilibrium behaviour, some follow information-sensitive behaviour, and only in special managed-routing settings do you get something close to system-optimal guidance. That maps very naturally to your two-chain idea: the "sat-nav" chain should be parameterised as a conditional-guidance class with compliance and adoption parameters, not as universal congestion minimisation. ²⁹

The fixed-point computation I would borrow is either a day-to-day dynamic traffic assignment loop with method-of-successive-averages style flow updates, or a mixed-equilibrium complementarity formulation when you explicitly separate guided and unguided populations. In practice, for your thesis, that probably means simulating several adoption rates and compliance levels, then fitting a destination-conditioned "guided" chain under those scenarios rather than assuming a single perfectly rational rerouting class. ³⁰

My implementation judgement is: adding partial-adoption and compliance parameters is **medium** difficulty and **high** realism payoff; building a full mixed-equilibrium inner loop is **high** difficulty and **medium** direct AUC payoff, but it materially strengthens the scientific credibility of the simulator-generated labels. Cross-cutting point: this topic feeds directly into **F**, because if the simulator's route-guidance logic is too deterministic, the inference problem becomes much easier than the real one and your validation becomes optimistic. ³¹

Microsimulation validation with SUMO

Papers I would keep closest to hand. Hesham Rakha, Brent Hellinga, Michel Van Aerde, and William Perez, "Systematic Verification, Validation and Calibration of Traffic Simulation Models," *TRB Annual Meeting paper*, 1996. This is still the clearest framework paper on separating verification, calibration, and validation and on not treating the simulator as self-justifying. ³²

Byungkyu Park and Hongtu Qi, "Development and Evaluation of a Procedure for the Calibration of Simulation Models," *Transportation Research Record* 1934, 208–217, 2005. This is the practical protocol paper I would actually imitate for staged calibration and multi-measure validation. ³³

Biagio Ciuffo and Carlos Lima Azevedo, “A Sensitivity-Analysis-Based Approach for the Calibration of Traffic Simulation Models,” *IEEE Transactions on Intelligent Transportation Systems* 15, 1298–1309, 2014, DOI 10.1109/TITS.2014.2302674. This is the strongest paper I found for reducing the calibration search space before you optimise anything. ³⁴

Pablo Alvarez Lopez et al., “Microscopic Traffic Simulation using SUMO,” *IEEE Intelligent Transportation Systems Conference*, 2018. This is the key SUMO reference: it positions SUMO as a scenario-generation and evaluation framework, not as automatic ground truth. ³⁵

The protocol implication for your thesis is blunt: you should not validate the inference method only against whatever SUMO emits, because SUMO’s outputs inherit all the modelling assumptions used to build the scenario. A credible protocol calibrates the simulator on one set of observables, validates it on held-out observables and held-out scenarios, and then evaluates the inverse method under multiple demand seeds and route-choice settings. For your use case, that means calibrating and validating not only counts and travel times, but also route shares, turning ratios, queue lengths, and if possible link-to-link transition frequencies, because those are closer to the target of your inverse model. The sensitivity-analysis literature is directly relevant because large microsimulators have too many parameters to calibrate all at once without overfitting. ³⁶

For SUMO specifically, the quirks you mentioned are not cosmetic. Official documentation says that vehicles are teleported when waiting exceeds a threshold, 300 seconds by default, as a gridlock-resolution device; routing in SUMO and duarouter is by shortest/fastest path on edge weights unless you introduce explicit stochasticity, and runs with the same inputs remain deterministic unless you vary the seed or activate randomisation. Those features are useful for simulation, but they are dangerous if you quietly treat the resulting trajectories as behavioural truth. In your experiments I would therefore log teleports at the trajectory level, either drop or separately analyse teleported trips, and treat routing randomness and rerouting settings as experimental factors rather than hidden defaults. ³⁷

My implementation judgement is: better validation protocol is **low-to-medium** difficulty and **medium** direct AUC payoff, but **very high** thesis-quality payoff. Cross-cutting point: van Oijen’s partial-sensor recursive-logit paper from topic **B** suggests exactly the kind of dual validation you want here—predict both behavioural trajectories and network flow consequences. ³⁸

Identifiability and uncertainty

Papers I would keep closest to hand. F. Barsotti, Y. de Castro, T. Espinasse, and P. Rochet, “Estimating the transition matrix of a Markov chain observed at random times,” 2014 preprint. This is the closest paper I found to your sparse-observation inverse setting; its key result is that censored Markov-chain inference is not identifiable without additional structure, and known sparsity/support restrictions are what rescue identifiability. ³⁹

Randal Douc, François Roueff, and Tepmony Sim, “The maximizing set of the asymptotic normalized log-likelihood for partially observed Markov chains,” *Annals of Applied Probability*, 2016; arXiv:1509.09048. This is the most useful general identifiability paper I found for partially observed Markov models: under weak conditions the likelihood identifies an *equivalence class* through invariant-distribution uniqueness, not necessarily a single parameter point. ⁴⁰

David Hsu, Aryeh Kontorovich, David A. Levin, Yuval Peres, Csaba Szepesvári, and Geoffrey Wolfer, “Mixing Time Estimation in Reversible Markov Chains from a Single Sample Path,” 2019. This is the best downstream-

UQ paper I retrieved for turning finite data into empirical intervals on stationary probabilities and spectral-gap type functionals. ⁴¹

Linan Wan et al., “Confidence intervals for Markov chain transition probabilities based on next generation sequencing reads data,” 2020, DOI 10.1007/s40484-020-0200-y. Despite the genomics application, this is directly relevant to your “many short trajectories” setting and gives asymptotic interval constructions for transition quantities under fragmented observations. ⁴²

The strongest conclusion here is that I did **not** find a published identifiability result tailored to the exact Morimura f , g inverse problem. The closest direct guidance is more general: partial-observation Markov problems are usually identifiable only after you impose structural constraints such as known zeros, equivalence-class restrictions, or low-dimensional parametrisations. That means your 8000+-parameter, 25%-observed setup is very unlikely to support edgewise certainty. The right target of inference is therefore not “every parameter is identified”, but “the specific predictive functionals I care about—trajectory likelihoods, class scores, perhaps some route-share summaries—are stable under the admissible uncertainty set”. ⁴³

Practically, I would do three things. First, quantify identifiability by inspecting the empirical Fisher or Hessian spectrum of the fitted model and by profiling a few scientifically meaningful functionals rather than raw coefficients. Secondly, report uncertainty on downstream prediction, not just on parameters: bootstrap vehicles and simulator seeds, and if you adopt Bayesian smoothing from topic **C**, propagate posterior draws of P through the hitting-probability equations and the trajectory detector. Thirdly, where possible, compress the model space with support constraints, destination conditioning, or latent classes rather than free per-edge effects, because those structural restrictions do double duty: they improve fit and make the inverse problem better posed. ⁴⁴

My implementation judgement is: uncertainty intervals on downstream AUC, LLR, and route predictions are **medium** difficulty and **high** thesis payoff; formal edgewise confidence intervals for all parameters are **high** difficulty and, in your current model, probably low value. Cross-cutting point: the same Fisher-information ideas that matter for topic **A** are also the right lens for “practical identifiability” here, and the same Bayesian priors from topic **C** provide the easiest route to uncertainty on g and on the detector itself. ⁴⁵

Top five techniques to try

First: replace the local+ ω -global softmax with a destination-conditioned recursive logit backbone.

Source papers: Fosgerau–Frejinger–Karlström 2013 and van Oijen–Daamen–Hoogendoorn 2020. What I would implement is a link-based recursive-logit model whose continuation value depends on destination, with your existing edge features entering local utility. If destinations are known in simulation, condition on them directly; if not, score over a small candidate-destination set. Experimental confirmation would be a clear rise in held-out trajectory log-likelihood and a narrowing of the gap to the empirical one-step ceiling, especially on the larger network where the present misspecification gap is biggest. Effort is **medium**; expected AUC gain is **high**. ⁴⁶

Second: restore Fisher-aware optimisation, ideally as natural-gradient or Fisher-scored Gauss–Newton, and wrap it in basin averaging. Source papers: Hayashi–Watanabe 2016, Hayashi 2022, Transtrum–Sethna 2012. What I would implement is a rowwise Fisher preconditioner for the softmax blocks, a damped Gauss–Newton/LM step on your log-residual vector, and then 10–20 multi-start fits with validation selection or predictive averaging. Experimental confirmation would be reduced seed-to-seed AUC variance, reduced basin dependence at fixed training loss, and better correlation between

training objective and held-out detection performance. Effort is **medium**; expected AUC gain is **medium-to-high**, with a particularly strong payoff in robustness. ⁵

Third: replace the hard floor in g with empirical-Bayes shrinkage and, if possible, posterior-derived hitting probabilities. Source papers: Alyousifi et al. 2021, Alyousifi et al. 2019, Assoudou-Essebbar 2003. What I would implement first is a Beta-Binomial / Dirichlet empirical-Bayes smoother for sparse pair-hit cells; the stronger version is to smooth P , derive posterior samples of g , and use posterior means or predictive scores in the loss. Experimental confirmation would be that the fitted_{fg} variant no longer degrades when pair counts are sparse, that estimated g -tables stop collapsing to the floor, and that adding g improves or at least stabilises AUC rather than merely increasing variance. Effort is **low-to-medium**; expected AUC gain is **medium** and the stability gain is likely larger than the raw accuracy gain. ¹⁶

Fourth: add a small latent mixture layer for behavioural heterogeneity. Source papers: Gupta-Kumar-Vassilvitskii 2016, Kausik-Tan-Tewari 2023, and for route-choice estimation scale, Mai et al.'s mixed recursive-logit decomposition method. What I would implement is either a two-to-four-component mixture of first-order chains or a mixed recursive-logit model with random coefficients for the "intrinsic" and/or "sat-nav" classes. Experimental confirmation would be higher held-out classification accuracy together with interpretable components, and evidence that different latent components align with plausible driver subtypes or compliance levels rather than arbitrary optimiser basins. Effort is **high**; expected AUC gain is **high but uncertain**, so this is not the first thing I would do, but it is the best next model-family extension after recursive logit. ⁴⁷

Fifth: replace the single point-estimate LLR by a model-averaged predictive score, with changepoint scoring as the fallback if rerouting is episodic. Source papers: Fitzpatrick-Stewart 2022, Arnesen-Holsclaw-Smyth 2015, and Görnitz et al. 2015. What I would implement is a predictive-density ratio averaged over multiple plausible fitted models; if rerouting happens mid-trip rather than trip-wide, I would compare that to a changepoint Markov score. Experimental confirmation would be improved calibration of class probabilities, better robustness to optimiser basin choice, and higher AUC specifically on trajectories that currently sit near the decision boundary. Effort is **medium**; expected AUC gain is **medium**, but the statistical defensibility gain is large because you stop pretending the fitted point estimates are unique. ⁴⁸

If I had to prioritise very aggressively, I would do these in this order: **destination-conditioned recursive logit**, **Fisher-aware optimisation with basin averaging**, **Bayesian shrinkage for g** , and only then **latent mixtures**. That ordering reflects my central judgement from the literature: your present bottleneck is more model-family misspecification than optimiser choice, but the optimiser pathology is real enough that it can easily mask improvements if left untreated. ⁴⁹

¹ ² ⁵ ⁴⁵ <https://projecteuclid.org/journals/annals-of-statistics/volume-44/issue-4/Information-geometry-approach-to-parameter-estimation-in-Markov-chains/10.1214/15-AOS1420.pdf>

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